

Question Number	Answer	Additional guidance	Mark
4(a)	An answer that includes two of the following points: • {outer / single} cell layer / trophoblast (1) • inner cell mass / embryoblast / contains pluripotent cells (on the inside) (1) • hollow / fluid filled / (fluid filled) cavity / blastocoel (1)	ignore totipotent ignore totipotent	(2)

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4(b)	An answer that includes the following: • calculation of number of mitotic cells (1)	<u>Example of calculation:</u> $= (15 \div 100) \times 60 = 9$	(1)

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Number			
4 (c)	An explanation that includes five of the following points: • (DNA methylation is the) addition of {methyl / CH₃} group(s) (1) • to cytosine (1) • (histone modification is the) {methylation / phosphorylation/ acetylation} of {histones / lysine} (1) • (resulting in) compacting chromatin / DNA wrapped more tightly around histones (1) • (resulting in) inhibition of {transcription factor / RNA polymerase} binding (1) • (therefore) preventing {transcription / gene expression} (1)	ACCEPT addition of methyl base ACCEPT CpG site / between cytosine and guanine ignore cysteine / genes ACCEPT ubiquitylation / sumoylation ACCEPT {adding / removing} of acetyl groups {in histone modification / to histones / to lysine} ACCEPT (histone modification results in) {euchromatin / heterochromatin / supercoiling / (highly) condensed DNA} Accept converse for acetylation e.g. loosening supercoiling ACCEPT converse for acetylation only ACCEPT silencing of gene / preventing mRNA production ACCEPT acetylation can allow {activation / transcription / expression} of gene	(5)